# Homework week 9


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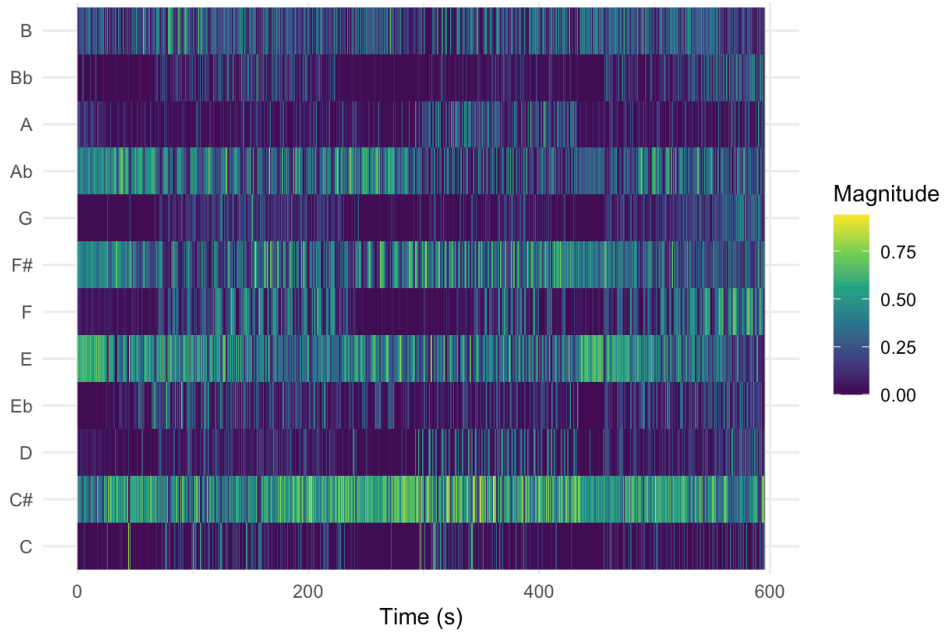
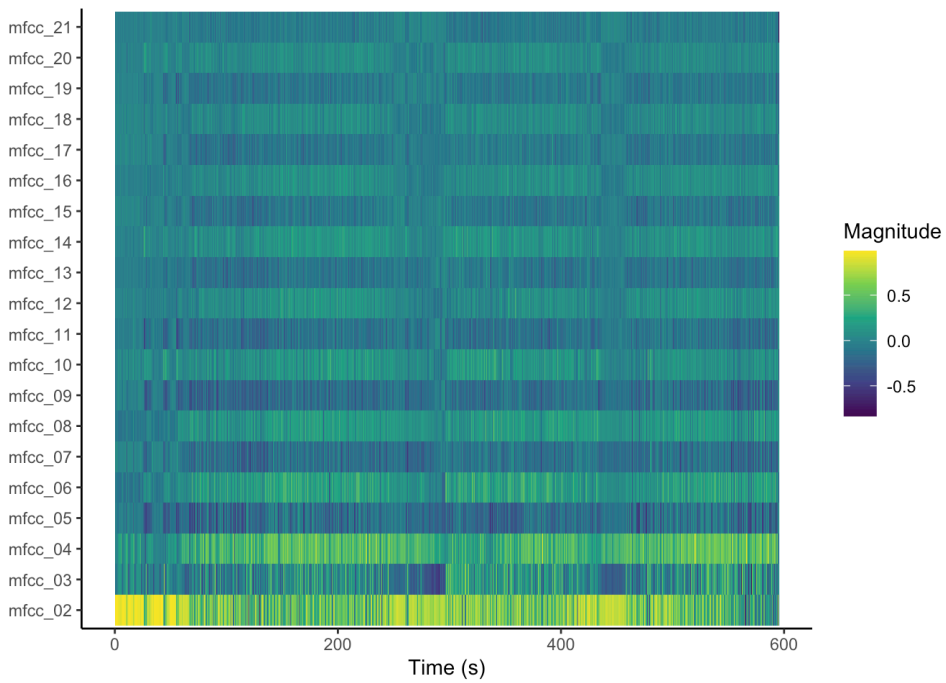
The images in this assignment are all based on the song ‘Words 2B Heard Meets Planetary Funk Alert’ by MC Conrad with Seba. The song is a groovy, chill drum and bass song.

## Chromagram:



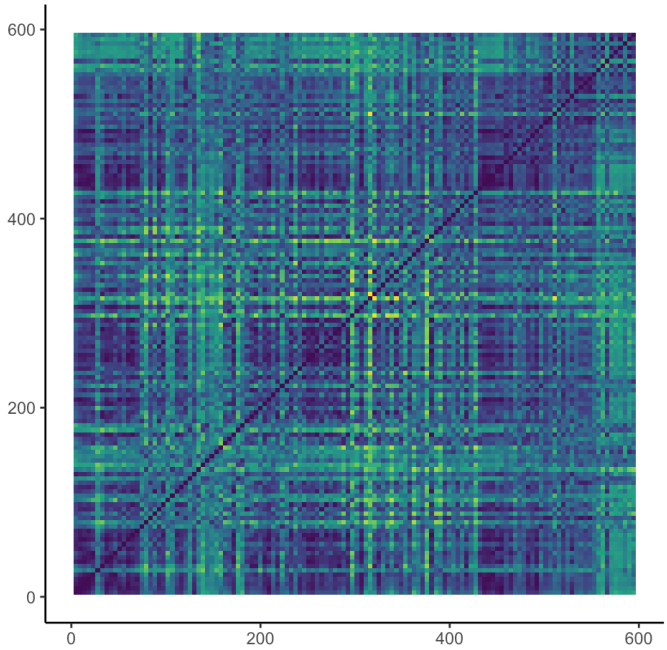

## Cepstrogram:



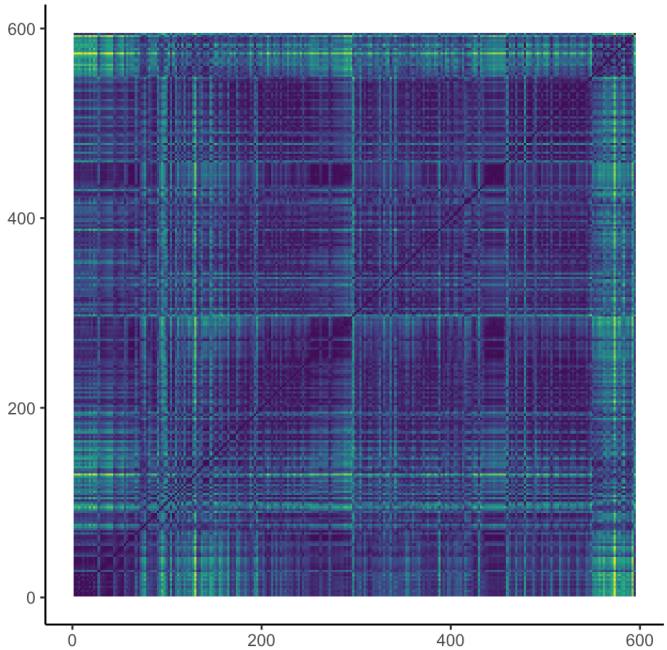

The clearest cepstrogram was produced by using the Euclidean normalization. No obvious pattern can be seen but most of the rows split into two or three distinct parts, which reflect the song structure.

# self-similarity matrices

## based on pitch:




## based on timbre:




For the self-similarity matrices, the Manhattan normalization was used. In both of them, but more in the pitch-based matrix, three parts can clearly be seen. However, I would describe the matrices as showing two distinct parts that feature ‘interruptions’ in them. This shows the songs two main grooves which feature calmer, buildup moments in them. Those moments can be seen in the lighter parts of the matrices. The argument of there being three parts rather than two can be made because the ‘interruption’ in the second part is distinct enough in the first image to show where it more or less starts and ends. In the second matrix that moment ends at the point with the darkest area (around 450) which shows something interesting happens there timbre-wise as well.